

Assessing Temporal Shifts in U.S. Diabetes Prevalence: A Stratified Time-Series Analysis, 2013-2023

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Abstract

Diabetes is a major contributor to morbidity, premature mortality, and rising health care costs in the United States, and its prevalence continues to increase nationwide [1]. National estimates, however, often mask important subgroup difference by age, sex, and geographic region, limiting the ability to design targeted prevention efforts. This study examines temporal trends in diabetes prevalence from 2013-2023 using data from the Behavioral Risk Factor Surveillance System (BRFSS), a nationally representative, annually administered survey of U.S. adults conducted by the Centers for Disease Control and Prevention (CDC) [2]. Survey-weighted prevalence estimates will be calculated for each year, stratified jointly by age group, sex, and region (Northeast, Southeast, Midwest, Southwest, West). Time-series visualizations and survey-weighted regression models will be used to assess whether disparities across age, sex, and region have widened or narrowed over time. I expect to observe persistently higher diabetes prevalence in older age groups, in men compared to women, and in the South and Midwest relative to other regions, with some evidence that the magnitude of these disparities changes over the decade. By integrating age, sex, and regional stratification into a single analytic framework, this study provides a more nuanced understanding of diabetes burden in the U.S., generating evidence to support geographically tailored and demographically informed prevention strategies.

Keywords: Diabetes prevalence; Spatiotemporal trends; BRFSS; Survey-weighted analysis; Age and sex disparities; Geographic variation

1. Introduction

Diabetes mellitus remains one of the most emergent public health challenges in the United States, contributing substantially to morbidity, premature mortality, and rising health care expenditures [1]. More than 38 million U.S. adults, accounting for approximately 11.6% of the population, are currently living with diabetes [2]. The clinical and economic consequences of diabetes extend beyond glycemic control, encompassing increased risks of cardiovascular disease, kidney disease, and diminished quality of life [3]. Over the past two decades, population-level diabetes prevalence has continued to rise, driven by trends in obesity, demographics, and persistent inequities in socioeconomic status and healthcare access [4,5]. Understanding how these burdens are distributed across demographic and regional subgroups is critical for informing targeted prevention and intervention strategies.

National surveillance systems provide essential insights into chronic disease trends, but aggregated estimates can obscure meaningful differences across populations. Prior research has documented prominent disparities in diabetes risk and outcomes by sex, age, race/ethnicity, and geography [5]. Sex related occurrences can occur due to differences in visceral adiposity,

healthcare utilization, and cardiometabolic risk behaviors [6]. Age is also a strong predictor of diabetes, with prevalence increasing sharply across adulthood [7]. Regional disparities further complicate trend variations, with the South and parts of the Midwest being labeled as the “diabetes belt”, with notorious disproportionately higher burdens of type 2 diabetes, shaped by the intersection of historical, structural, and environmental determinants of health [8]. Despite clear evidence of these disparities, relatively little work has examined how these patterns evolve over time, especially using a unified, survey-weighted framework that supports nationally representative comparisons.

The Behavioral Risk Factor Surveillance System (BRFSS), administered annually by the Centers for Disease Control and Prevention (CDC), is the largest continuously conducted health survey in the world and provides extensive data on health behaviors, chronic conditions, and social determinants of health among U.S. adults [9]. BRFSS’s complex probability sampling design enables the generation of reliable, population-weighted estimates at both national and regional levels. The survey includes standardized questions on physical-diagnosed diabetes, demographic characteristics, and state of residence, making it an optimal data source for spatiotemporal analyses of diabetes prevalence. This data source also supports the evaluation of long-term trends and the identification of subpopulation differences that might not be visible in shorter or more limited datasets.

Despite the utility of BRFSS for chronic disease surveillance, few studies have jointly examined how diabetes prevalence has evolved across age groups, sex, and U.S. regions over a decade-long period. Combining these three dimensions is essential to assess how trends may differ in magnitude or direction across subgroups, with important implication for public health planning. A comprehensive, stratified trend analysis can therefore enlighten nuanced population patterns that are masked in aggregated national estimates.

The present study addresses this gap by examining temporal trends in diabetes prevalence across U.S. adults from 2013 to 2023, stratified simultaneously by age group, sex, and geographic region using nationally representative BRFSS data. This analysis quantifies annual, survey-weighted diabetes prevalence for each subgroup and evaluates whether temporal trajectories differ across these intersecting demographic dimensions. Time-series visualizations and survey-weighted regression models are used to assess whether disparities have persisted, widened, or narrowed over the study period. By integrating multiple demographic and geographic factors, this study aims to generate a detailed, policy-relevant understanding of the evolving distribution of diabetes burden in the United States.

Ultimately, this work seeks to inform more targeted and equitable diabetes prevention efforts by identifying where and among whom the greatest burdens and fastest changes are occurring. The findings will provide timely evidence to guide regional public health planning, age-specific intervention strategies, and sex-informed approaches to chronic disease prevention and management.

2. Data and Methods

2.1. Study Design

This study uses a repeated cross-sectional design to examine temporal and demographic patterns in diabetes prevalence across the United States from 2013-2023. Data were obtained from the Behavioral Risk Factor Surveillance System (BRFSS), an annual, nationally representative telephone survey administered by the CDC in collaboration with state health departments. The BRFSS collects health-related information from non-institutionalized adults aged ≥ 18 years across all 50 states, the District of Columbia, and U.S. territories, making it the largest continuously conducted health survey in the world [9]. Because BRFSS samples a new set of respondents each year, its structure is inherently cross-sectional, but its annual periodicity enables assembly of a population-level time series for monitoring secular disease trends.

The study population consists of all adult BRFSS respondents from survey years 2013 through 2023. These years were selected to align with both the CDC’s post-2011 weighting methodology

(“ranking”) and the availability of harmonized variables on diabetes status, sex, age, and census region. The spatial domain encompasses all U.S. census division aggregated into five analytic regions (Northeast, Southeast, Midwest, Southwest, and West), selected to improve interpretability and align with well-documented geographic disparities in chronic disease burden. Survey year serves as the time index, enabling examination of linear and subgroup-specific temporal trends.

This design allows for the estimation of annual, survey-weighted prevalence of diabetes across sex, age groups, and regions. This provides insights into population level changes over time while recognizing the cross-sectional nature of the underlying study

2.2. Data Sources

BRFSS is collected through computer-assisted telephone interviewing (landline and cellular), using a complex sampling design involving stratification, clustering, and weighting to achieve national and state representativeness [10]. The survey’s mission is to track chronic conditions, health behaviors, and risk factors to support public health surveillance and inform policy. All data used in this study were publicly accessible through the CDC BRFSS website and downloaded as a SAS transport or dataset files for each survey year.

Each annual BRFSS dataset includes over 400,000 respondents, with sampling weights that account for unequal probabilities of selection and nonresponse [10]. This weighting is fundamental for producing unbiased population estimates, particularly for outcomes such as diabetes that vary widely across demographic groups.

Data cleaning and processing steps began with importing the annual BRFSS datasets into SAS from each year and selecting analytic variables including diabetes status, sex, age, region, survey year, and sampling weights. Variables were recoded to ensure consistency across years. Diabetes status and sex were recoded as binary variables, while age was categorized into four groups (18-34, 35-54, 55-64, and 65+) and region collapsed into five groups based on respondent state. All missing or invalid responses were excluded from the analysis. Before computing prevalence estimates, survey weights were applied to obtain weighted prevalence and 95% confidence intervals. For each year, weighted prevalence of diabetes was estimated within each sex x age group x region stratum. Annual prevalence estimates were combined into a single analytic dataset.

All data management, weighting, and statistical analyses were conducted using SAS version 9.4, which is well equipped for handling large, complex survey datasets and supports appropriate variance estimation for weighted analyses [11].

2.3. Outcomes and Predictors

The primary outcome was survey-weighted self-reported diabetes prevalence, defined as the proportion of adults reporting a provider diagnosis of diabetes (excluding gestational diabetes). Although coded as a binary variable in the raw dataset, the analyses used annual weighted prevalence percentages as the dependent variable for trend modeling.

Primary predictors included year, sex, age group, and U.S. census region. Year was treated as a continuous variable and centered prior to modeling ($\text{year_c} = \text{year} - 2013$) to reduce numerical scaling issues in regression estimation. Sex was coded as male or female. Age group was categorized into four groups, 18-34, 35-54, 55-64, 65+, balancing the distribution of participants evenly between groups. Region was categorized into Northeast, Southeast, Midwest, Southwest, West. No transformations were applied given the categorical structure and the use of survey-weighted prevalence estimates rather than individual-level continuous outcomes.

2.4. Data Analysis

The data analysis strategy consisted of three sequential components: (1) constructing weighted annual estimates of diabetes prevalence for every combination of sex, age group, and region; (2) fitting regression models to quantify temporal trends; and (3) visualizing subgroup-specific patterns using structured time-series plots.

For each survey year, sex- and age- specific diabetes prevalence rates were estimated accounting for BRFSS sampling weights, strata, and primary sampling units. Year- specific estimates were then combined across survey years to produce a panel dataset for regression modeling.

Considering BRFSS provides annual cross-sections rather than longitudinal follow-up, models were fit using ordinary least squares regression on annual weighted prevalence percentages, not individual data.

To evaluate how diabetes prevalence changed over time and how these changes differed across demographic groups, four nested regression models were estimated:

The core model was:

$$\mu(\text{Prevalence}) = \beta_0 + \beta_1(\text{Year}_c) + \epsilon$$

where β_1 represents the annual percentage-point change in diabetes prevalence and year_c represents the centered year variable.

To evaluate subgroup-specific trajectories, models were expanded to include categorical predictors:

$$\text{Model 2} - \mu(\text{Prevalence}) = \beta_0 + \beta_1(\text{Year}_c) + \beta_2(\text{Sex}) + \epsilon$$

$$\text{Model 3} - \mu(\text{Prevalence}) = \beta_0 + \beta_1(\text{Year}_c) + \beta_2(\text{Sex}) + \beta_3(\text{AgeGroup}) + \epsilon$$

$$\text{Model 4} - \mu(\text{Prevalence}) = \beta_0 + \beta_1(\text{Year}_c) + \beta_2(\text{Sex}) + \beta_3(\text{AgeGroup}) + \beta_4(\text{Region}) + \epsilon$$

All categorical predictors were dummy coded using women, ages 18-34, and Northeast as reference groups. Model 1 estimates the overall temporal trend. Models 2-4 quantify how this trend persists after accounting for demographic structure.

Because the analytic goal was to characterize population-level trends rather than estimate causal relationships, additional adjustment for factors such as race/ethnicity, socioeconomic status, or BMI was not undertaken. Adjusting for these variables may obscure true population trends if they lie on the causal pathway or reflect legitimate demographic changes over time.

Model diagnostics included examination of residual plots, Q-Q plots, and tests for linearity and homoscedasticity. Centering survey year improved numerical stability and interpretability of model intercepts. Diagnostics supported the adequacy of linear trend models for summarizing annual prevalence changes.

Visualizations were designed to convey trends clearly across demographic and geographic dimensions. The primary figure consists of a four-panel time-series plot (one panel per age group) displaying male and female trend lines with 95% confidence intervals. After identifying the age group with the highest prevalence, a follow-up visualization stratified that age group by U.S. Census region, illustrating regional differences in temporal patterns.

3. Results

3.1. Overall Trends in Diabetes Prevalence, 2013-2023

3.1.1. Changes over Time

Across all models, diabetes prevalence increased steadily between 2013 and 2023. In the simplest model (Model 1), year was positively associated with diabetes prevalence ($\beta = 0.097$, SE = 0.0105; Table 1). Although this coefficient was not statistically significant in Model 1, the direction and magnitude were consistent across increasingly adjusted models. After adjusting for sex, age group, and region, the coefficient for year became highly precise and statistically significant ($\beta = 0.097$, SE 0.018, $p < 0.001$), indicating that prevalence increased by approximately 0.10 percentage points per year. This pattern is reflected visually in Figures 1-3, where trends consistently show upward trajectories across demographic groups.

Table 1. Linear Regression Models Predicting Diabetes Prevalence Over Time, BRFSS 2013-2023.

Model	Year	Sex	Age Group			Region				Intercept
		Male	35-49	50-64	≥ 65	Midwest	Southeast	Southwest	West	
1	0.097 (0.105)	-	-	-	-	-	-	-	-	12.41 (0.54)***
2	0.097 (0.104)	1.95 (0.69)**	-	-	-	-	-	-	-	11.43 (0.64)***
3	0.097 (0.023)***	1.95 (0.15)***	6.7 (0.21)***	16.2 (0.21)***	21.40 (0.21)***	-	-	-	-	0.36 (0.19)
4	0.097 (0.018)***	1.95 (0.12)***	6.70 (0.17)***	16.2 (0.17)***	21.4 (0.17)***	0.38 (0.19)*	2.19 (0.19)***	2.52 (0.19)***	0.06 (0.19)	-0.67 (0.19)

Values are unstandardized coefficients (SE). Reference groups: women, age 18-34, Northeast Region

* p < 0.05, ** p < 0.01, *** p < 0.001

3.2. Sex Differences in Diabetes Prevalence

3.2.1. Main Effects of Sex

Men consistently exhibited higher diabetes prevalence than women. In Model 2, adjusting only for year, men had an estimated 1.95-percentage point higher prevalence than women ($\beta = 1.95$, SE = 0.69, $p < 0.01$). This sex difference remained stable with additional adjustments in Models 3-4 ($\beta = 1.95$, SE = 0.12, $p < 0.001$ in Model 4), indicating a persistent effect.

3.2.2. Sex Differences Over Time

Figures 1a-d illustrate weighted diabetes prevalence stratified jointly by sex and age group. Across every age category, men displayed higher prevalence than women throughout the full 2013-2023 period. Notably:

- Among adults age 18-32, absolute prevalence remained low for both sexes (Figure 3a), but men maintained a small advantage of ~1%.
- In the 35-54 and 55-64 age groups (Figures 1b-c), the sex gap widened, particularly after 2018 when men showed steeper increases.
- In adults 65 years and older (Figure 1d), prevalence was highest overall, and the absolute sex difference persisted across the decade.

Figure 1a-d: Weighted Diabetes Prevalence by Sex and Age Group, BRFSS 2013-2023

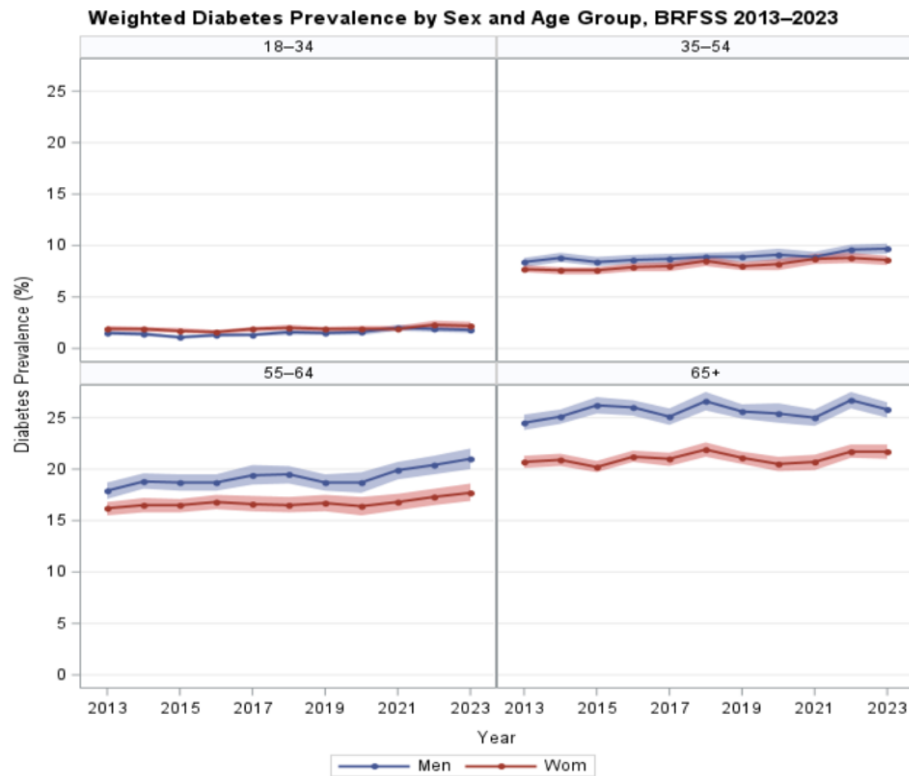


Figure 1. Weighted diabetes prevalence by sex and age group from 2013–2023. These descriptive estimates reflect survey weighting from the BRFSS complex sampling design. Across all age groups, men have consistently higher prevalence than women, and prevalence increases steadily with age. (a) Upper left panel. Weighted diabetes prevalence by sex in the 18–34 category. (b) Upper right panel. Weighted diabetes prevalence by sex in the 35–54 category. (c) Bottom left panel. Weighted diabetes prevalence by sex in the 55–64 category. (d) Bottom right panel. Weighted diabetes prevalence by sex in the 65+ category.

3.3. Age-Group Differences in Diabetes Prevalence

3.3.1 Cross-Sectional Age Differences

Adjusting for year and sex (Model 3), diabetes prevalence increased dramatically with age (Table 1):

- Ages 35–54: Increased by 6.7 percentage points (SE = 0.21, $p < 0.001$)
- Ages 55–64: Increased by 16.2 percentage points (SE = 0.21, $p < 0.001$)
- Ages ≥ 65 : Increased by 21.4 percentage points (SE = 0.21, $p < 0.001$)

Relative to adults aged 18–34, each successive age group showed substantially higher prevalence, confirming that age is the strongest demographic predictor in the model.

3.3.2. Age-Specific Trends Over Time

Predicted values from Model 4 (Figure 2) show minimal divergence in slopes across age groups. All groups exhibited parallel increases of approximately 0.20 percentage points per year,

indicating the rise in prevalence is happening across the full adult age spectrum. Despite similar slopes, absolute levels differed notably:

- Adults ≥ 65 years had the highest predicted prevalence (~22-24%).
- Adults 55-64 followed, increasing from ~17.5% to ~18.5%.
- Adults 35-54 remained in the mid ~8% to 9% range.
- Adults 18-34 exhibited consistently low levels (<3%).

Figure 2: Predicted Diabetes Prevalence Over Time by Age Group

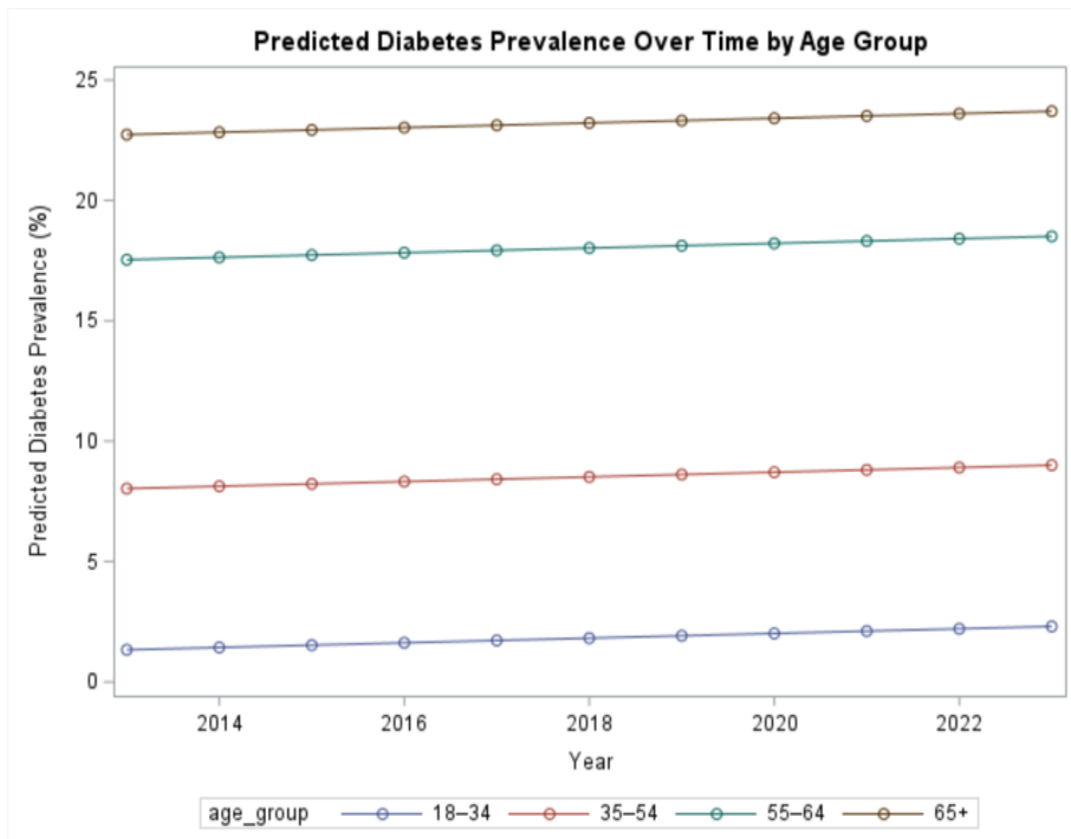


Figure 2. Predicted diabetes prevalence from 2013 to 2023 by age group, estimated using Model 4 (adjusted for year, sex, region, and age group). Prevalence increases with age, with adults aged 65 and older showing the highest burden. All age groups exhibit parallel upward trends of approximately 0.10 percentage points per year.

3.4. Regional Differences in Diabetes Prevalence

3.4.1. Cross-Sectional Regional Differences

After adjusting for year, sex, and age group, several regions had significantly higher diabetes prevalence compared with the Northeast (reference group) as estimated in Model 4;

- Southeast: Increase in 2.19 percentage points (SE = 0.19, $p < 0.001$)
- Southwest: Increase in 2.52 percentage points (SE = 0.19, $p < 0.001$)

- Midwest: Increase in 0.38 percentage points (SE = 0.19, $p < 0.05$)
- West: Not significantly different ($\beta = 0.06$, SE = 0.19, $p = 0.76$)

These results highlight a clear geographic gradient, where the Southeast and Southwest regions carry disproportionately greater burdens.

3.4.2. Region-Specific Predicted Trends

Figure 3 shows predicted diabetes prevalence across regions using Model 4. Patterns indicate:

- The Southwest and Southeast consistently exhibit the highest predicted prevalence (~13.8%-14.8% over the decade).
- The Midwest forms a middle tier (~11.7-12.7%).
- The Northeast and West remain the lowest-burden regions (~11.4-12.4%).

These regional differences align with patterns documented in prior epidemiologic research, and underscore the importance of structural determinants of health.

Figure 3: Predicted Diabetes Prevalence Over Time by Region

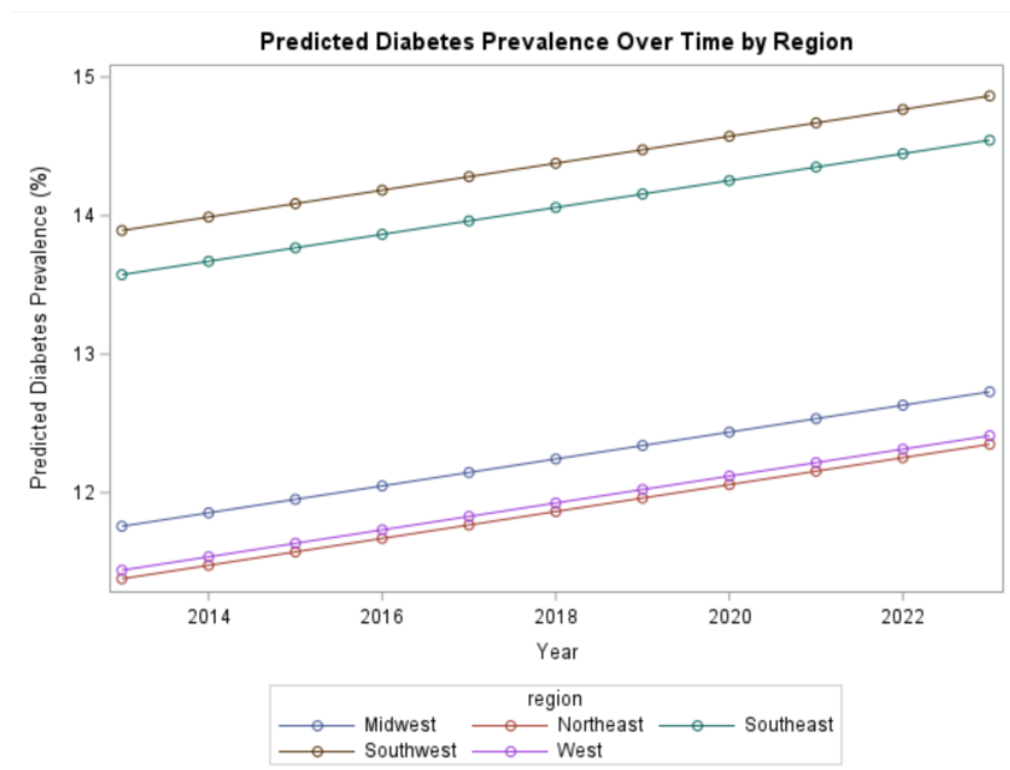


Figure 3. Predicted diabetes prevalence from 2013 to 2023 by U.S. region, based on Model 4 (adjusted for year, sex, age group, and region). Predicted values were generated from the fully adjusted linear regression model. The southeast and Southwest regions consistently show the highest prevalence, while the Northeast and West remain lowest across the period.

Across all models, the coefficient for year remained identical in magnitude ($\beta = 0.097$). The narrowing of the standard error as covariates were added suggests that demographic factors

explain some variability. Collectively, figures 1 through 3 triangulate the regression findings, showing coherence between model-based predictions and empirical sample estimates.

4. Discussion

This study examined national trends in diagnosed diabetes prevalence in the United States from 2013 to 2023, using repeated cross-sectional data from BRFSS. By combining descriptive weighted estimates with regression-based predictions stratified by sex, age group, and geographic region, this analysis provides a contemporary, population-representative assessment of how the diabetes burden has evolved over the past decade. Several consistent patterns emerged across models and visualizations. First, diabetes prevalence remained substantially higher among men than women across all years and analytic approaches. Second, prevalence rose sharply with age, with adults aged ≥ 65 experiencing levels more than 20 percentage points higher than younger adults. Third, while regional differences were smaller in magnitude than age or sex differences, the Southeast and Southwest consistently showed the highest predicted prevalence. Across all models, the year coefficient was positive, small in magnitude, and highly stable across adjustments, indicating a slow, steady increase in diabetes prevalence over time,

The sex differences observed in this analysis align closely with existing epidemiologic literature. Men had an estimated 1.95 percentage point higher prevalence than women after adjustment, a difference that remained stable across all four models. These findings are consistent with prior evidence attributing sex differences to biologic factors such as greater visceral adiposity accumulation in men, as well as social and behavioral patterns [12]. These patterns include lower healthcare utilization, delayed diagnosis, and reduced engagement in chronic disease prevention and management behaviors. The persistence of the sex coefficient across models suggests that sex differences in diabetes risk remain robust and unlikely to be explained by confounding.

Age differences were pronounced and reflect well-established life-course patterns in metabolic risk [13]. Relative to adults aged 18-34, predicted prevalence was 6.7 percentage points higher among adults aged 35-54, 16.2 points higher among adults aged 55-64, and more than 21 points higher among adults ≥ 65 . These large differences reinforce decades of evidence showing that cumulative exposure to behavioral, metabolic, and environmental risk factors, and age-related changes in adiposity and insulin sensitivity contribute to sharply rising diabetes risk with age [13]. Despite substantial differences in baseline prevalence, all groups exhibited nearly identical time trends, suggesting that underlying risk trajectories are rising uniformly across the U.S. adult population, rather than disproportionately accelerating in any particular age group.

Regional results further illuminate the heterogeneity of diabetes risk across the United States. After adjustment for sex and age, Southeast, Southwest, and Midwest residents had higher predicted prevalence than adults living in the Northeast, the reference region. The coefficient for Southeast ($\beta = 2.19$), Southwest ($\beta = 2.52$), and Midwest ($\beta = 0.38$) indicate modest but meaningful regional disparities. These findings align with national reports documenting elevated obesity, lower physical activity, food insecurity, and structural barriers to healthcare primarily in the Southeast, though similar patterns are increasingly evident in parts of the Southwest and Midwest [14, 15]. Visualizations demonstrated that regional lines were approximately parallel, indicating that disparities have not widened or narrowed meaningfully over the decade, and rather, geographic inequalities remain persistent but stable.

Across all models, the estimated annual increase in diabetes prevalence was nearly identical ($\beta = 0.097$). The stability of this coefficient, even after adjustments, indicates that the gradual rise in diabetes prevalence is not attributable to demographic shifts in these characteristics. Instead, the results point toward underlying population-wide drivers of diabetes risk. The consistency of the

trend across models strengthens the conclusion that diabetes prevalence is increasing slowly but steadily in the U.S. adult population. Although the magnitude of this increase appears to be small (~0.10 percentage points per year), this equates to meaningful growth at the population level over a decade, particularly when combined with the already high prevalence in older adults and in certain regions.

This study benefits from the large sample size, national coverage, and repeated annual structure of BRFSS. The survey's weighting procedures enable population-level inference, and the combination of descriptive and model-based approaches allows for clearer interpretation of both absolute and relative patterns.

However, there are several limitations inherent to the study design that should be considered. As a repeated cross-sectional survey, BRFSS cannot track individual-level change, limiting the ability to determine whether prevalence trends are driven by cohort replacement, diagnostic changes, or shifts within individuals over time. Self-reported physician diagnosis may underestimate true prevalence, especially among populations with limited healthcare access or fewer screening opportunities. Regional estimates may also be influenced by state-level sample size differences, though weighting helps mitigate this. Additionally, although the BRFSS timeframe is adequate for detecting long-term linear trends, more complex temporal behaviors may remain undetected. Despite these limitations, BRFSS remains one of the most robust national surveillance systems for monitoring chronic disease prevalence, and its annual design is well-suited to detecting long-term trends and disparities.

Further research should explore racial/ethnic, socioeconomic, and policy-level contributors to the stable, yet persistent disparities observed here. Integrating environmental data, such as food environment indicators, built environment characteristics, and state policy measures, may clarify why certain regions consistently bear higher burdens. More complex modeling approaches could distinguish generational from temporal influences. Additionally, exploring non-linear trends or breakpoints may provide insight into whether recent public health or economic events have shifted diabetes trajectories.

5. Conclusions

This study provides an updated national assessment of diabetes prevalence trends in the United States from 2013-2023, incorporating demographic and regional comparisons using BRFSS data. Results show a small but consistent annual increase in diabetes prevalence, alongside persistent disparities by sex, age group, and geographic region. Men, older adults, and residents of the Southeast and Southwest continue to experience the highest burden. Regression models demonstrate stable, parallel time trends across all subgroups, indicating that while disparities are substantial, they are not widening.

Despite limitation inherent to self-report and repeated cross-sectional data, these findings highlight the importance of sustained, targeted prevention and intervention strategies. Continued surveillance is essential for monitoring progress and guiding public health policy, particularly in high-burden regions and populations.

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